$$\int_{-\ln 2}^{-\ln 10} \frac{e^x}{e^{2x}-1}\,dx \qquad e^x = y \quad e^x dx = dy \quad x = -\ln 10 \to y = e^{-\ln 10} = \frac{1}{10}$$

$$\int_{1/10}^{1/2} \frac{1}{y^2-1}\,dy \qquad \frac{1}{y^2-1} = \frac{1}{(y+1)(y-1)}$$

Condizione di AB, tolti dx: $\frac{A}{y+1} + \frac{B}{y-1}$

$$\frac{A(y-1)+B(y+1)}{y^2-1} = \frac{(A+B)y + (B-A)}{y^2-1} \qquad \begin{cases} A+B=0 \\ B-A=1 \end{cases} \qquad \begin{matrix} 2A+1=0 \\ B=A+1 \end{matrix} \qquad A=-\tfrac{1}{2} \quad B=\tfrac{1}{2}$$

$$\frac{1}{y^2-1} = \frac{-1/2}{y+1} + \frac{1/2}{y-1}$$

$$= \int_{1/10}^{1/2} \left(\frac{-1/2}{y+1} + \frac{1/2}{y-1}\right) dy =$$

$$= -\frac{1}{2}\left[\log|y+1|\right]_{1/10}^{1/2} + \frac{1}{2}\left[\log|y-1|\right]_{1/10}^{1/2}$$

$$\int_0^1 \frac{x^2+4x+5}{4x^2+4x+1}\,dx$$

$x^2+4x+5 \;\big|\; 4x^2+4x+1$

$x^2+x-1/4 \;\big|\; 1/4$

$3x+\frac{19}{4}$

$$x^2+4x+5 = \frac{1}{4}(4x^2+4x+1) + \left(3x+\frac{19}{4}\right)$$

$$= \int_0^1 \frac{1}{4}\,dx + \frac{1}{4}\int_0^1 \frac{12x+19}{4x^2+4x+1}\,dx$$

$$\frac{12x+19}{(2x+1)^2} \overset{?}{=} \frac{A}{2x+1} + \frac{B}{(2x+1)^2}$$

$$\frac{A(2x+1)+B}{(2x+1)^2} \to \frac{2Ax+(A+B)}{(2x+1)^2} \to \begin{cases} 2A=12 \\ A+B=19 \end{cases} \qquad A=6 \quad B=13$$

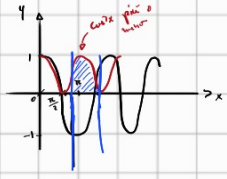
$$= \left[\frac{x}{4}\right]_0^1 + \frac{3}{4}\int_0^1 \frac{2}{2x+1}\,dx + \frac{1}{4}\int_0^1 \frac{13}{(2x+1)^2}\,dx$$

$$= \frac{1}{4} + \frac{3}{4}\left[\ln|2x+1|\right]_0^1 + \frac{13}{8}\int_0^1 \frac{2}{(2x+1)^2}\,dx =$$

$$= \frac{1}{4} + \frac{3}{4}\ln 3 + \frac{13}{8}\left[-\frac{1}{2x+1}\right]_0^1 =$$

$$= \frac{1}{4} + \frac{3}{4}\ln 3 + \frac{13}{8}\left(-\frac{1}{3}+1\right) = \frac{4}{3} + \frac{3}{4}\ln 3$$

③ Area di $R = \left\{(x,y)\in\mathbb{R}^2 \;:\; x\in\left[\frac{3}{4}\pi, \frac{7}{2}\pi\right],\; 0 \le y \le \cos^2 x\right\}$



$$\cos(2x) = \cos^2 x - \sin^2 x = \cos^2(x) - 1$$

$$\cos^2 x = \frac{\cos(2x)+1}{2}$$

$$\text{Area}(R) = \int_{\frac{3}{4}\pi}^{3\pi} \cos^2 x\,dx = \int_{\frac{3}{4}\pi}^{3\pi} \frac{\cos(2x)+1}{2}\,dx$$

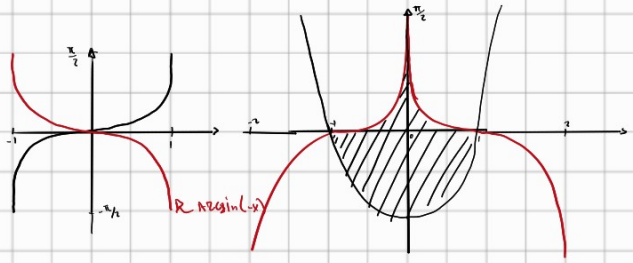
$$= \frac{1}{4}\int_{\frac{3}{4}\pi}^{3\pi} 2\cdot\cos(2x)\,dx + \frac{1}{2}\int_{\frac{3}{4}\pi}^{3\pi} 1\,dx =$$

$$= \frac{1}{4}\left[\sin(2x)\right]_{\frac{3}{2}\pi}^{3\pi} + \frac{1}{2}\left[x\right]_{\frac{3}{4}\pi}^{3\pi} = \frac{1}{4}\left(\sin(3\pi) - \sin\left(\tfrac{3}{2}\pi\right)\right) + \frac{1}{2}\left(\tfrac{3}{2}\pi - \tfrac{3}{4}\pi\right) =$$

$$= \frac{1}{4} + \frac{3}{8}\pi$$

④ Area di $B = \left\{(x,y)\in\mathbb{R}^2 \;:\; x^2-1 \le y \le \arcsin(1-|x|)\right\}$

$\sin\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1,1]$



$$\text{Area}(B) = \int_{-1}^{1} \left(\arcsin(1-|x|) - (x^2-1)\right) dx$$

$$= 2\int_0^1 \left(\arcsin(1-|x|) - x^2 + 1\right) dx$$

$$= 2\int_0^1 \arcsin(1-x)\,dx - 2\int_0^1 x^2\,dx + 2\int_0^1 dx =$$

$$= -2\int_1^0 \arcsin(y)\,dy - 2\left[\frac{x^3}{3}\right]_0^1 + 2\left[x\right]_0^1 \qquad \begin{matrix} 1-x=y \\ -dx=dy \end{matrix}$$

$$= 2\int_0^1 \arcsin(y)\,dy - \frac{2}{3} + 2 =$$

$$= 2\left[y\arcsin y\right]_0^1 - 2\int_0^1 y\frac{1}{\sqrt{1-y^2}}\,dy + \frac{4}{3} \qquad \begin{matrix} 1-y^2=t \\ -2y\,dy=dt \end{matrix}$$

$$= 2\arcsin(1) - \int_1^0 \frac{1}{\sqrt{t}}\,dt + \frac{4}{3}$$

$$= 2\frac{\pi}{2} + \left[\frac{t^{1/2}}{1/2}\right]_1^0 + \frac{4}{3} = \pi + \frac{4}{3} - 2 = \pi - \frac{2}{3}$$